

TECHNICAL EDUCATION AND TRAINING
TELANGANA
DIPLOMA EXAMINATION (C-21)
C-21-DEC-2025
SEMESTER V, SEMESTER END EXAM
CCB/CPS/CS/ES
CS-575/ES-574
Artificial Intelligence



PCOI
1504

Exam Date: 04-12-2025

Duration: 2 Hours [10:00 AM To 12:00
NOON]

Session:

[Total Marks:

PART-A

Instructions:

1. Answer the following questions.
2. Each question carries **ONE** mark.

8 X 1 = 8

1. Define First order logic.
2. What is Supervised Learning?
3. What is the impact of background noise on speech recognition systems?
4. List uninformed search strategies
5. List semantic analysis technique
6. What is the use of parser?
7. What are the benefits of AI-driven cars?
8. What are some of the key benefits of using AI in computer applications?

PART-B

Instructions:

1. Answer the following questions.
2. Each question carries **THREE** marks.

4 X 3 = 12

- (a). Write about search in State spaces
----- OR -----
- (b). Explain the steps involved in communication agent
- (a). List the difficulties in reasoning with uncertain information.
----- OR -----
- (b). **How can AI be used to improve the safety of robotics applications?**
- (a). List six advantages of Natural language processing
----- OR -----

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- 11(b). Explain Transition Networks 004
- 12(a). How is AI being used to improve healthcare accessibility and equity?
---- OR ----
- 12(b). Differentiate between a traditional drone and an AI-based drone

PART-C

Instructions:

1. Answer the following questions.
2. Each question carries **FIVE** marks.

4 X 5 = 20

- 13(a). Explain about state space graph 004
---- OR ----
- 13(b). Explain inflectional morphology with an example
- 14(a). Explain over fitting
---- OR ----
- 14(b). **Differentiate HMM, DNN, CNN models in speech recognition.**
- 15(a). Write the differences between semantics and pragmatic
---- OR ----
- 15(b). Explain different forms of ambiguity 004
- 16(a). Discuss in detail about the significance of AI in observing the universe.
---- OR ----
- 16(b). Explain the role of AI in military applications.